

18-661: Introduction to ML for Engineers

Practice Questions for MidTerm Exam

Spring 2026

ECE – Carnegie Mellon University

Question: MLE (Maximum Likelihood Estimation) and MAP (Maximum a Posteriori) estimation yield the same result when the prior distribution in MAP is uniform.

- ☐ True
- ☐ False

Maximum Likelihood Estimation

True: because in the case that the prior distribution in MAP is strictly uniform, the $P(\theta)$ is uniform across all θ .

Problem 3

Question: Suppose a dataset has features x_1 and x_2 and label y . Both Naive Bayes and Logistic Regression classifiers assume conditional independence of the features given the label, that is,

$$p(x_1, x_2|y) = p(x_1|y)p(x_2|y)$$

- ☐ True
- ☐ False

Problem 3

False: False. Naive Bayes classifier assumes that the features of the data are conditionally independent given the class label. But Logistic regression does not assume this.

- Logistic Regression:

$$P(D) = \prod_{n=1}^N p(y_n | x_n; w) = \prod_{n=1}^N \sigma(w^\top x_n)^{y_n} [1 - \sigma(w^\top x_n)]^{1-y_n}$$

- NB:

$$p(y | x) \propto p(y) \prod_{d=1}^D p(x_d | y).$$

Problem 4

Question: When the training dataset has a large number of samples and has high-dimensional features, the test-time computational complexity (in terms of the numbers of training samples and features) of K-Nearest Neighbors (KNN) is smaller than that of logistic regression.

- ☐ True
- ☐ False

Problem 4

False: KNNs struggle with large, high-dimensional datasets because they require computation of distances to each point in the training dataset.

Problem 5 (SVM)

Question: The dual formulation of SVM (support vector machines) will have fewer variables to optimize on a training dataset $\mathcal{D}_1$ with 1,000 data points with 20 features each than on a dataset $\mathcal{D}_2$ with 200 data points with 100 features each.

- ☐ True
- ☐ False

Problem 5 (SVM)

False: In the dual formulation, the number of variables to optimize is dependent on the number of data points (one Lagrange multiplier α_i per data point) but independent of the number of features.

- $\mathcal{D}_1$: 1,000 variables.
- $\mathcal{D}_2$: 200 variables.

Problem 6 (Bias-Variance)

Question Part A: Suppose you are given a linear regression problem. You decide to rescale the features to be zero-centered and have unit standard deviation. Then you find a new model that fits the rescaled data.

The bias component of the expected risk of the new model with feature scaling is strictly smaller than that of the old model without feature scaling.

- ☐ True
- ☐ False

Problem 6 (Bias-Variance)

False: Feature scaling (normalization/standardization) helps with the optimization process (gradient descent convergence) but does not change the capacity of the Linear Regression model or the space of functions it can learn. The bias remains the same.

Problem 6 (Bias-Variance)

Question Part B: You collect more data, keep the original features, and learn a new model. Then the variance component of expected risk for the new model is smaller or equal to the variance of the old model.

- ☐ True
- ☐ False

Problem 6 (Bias-Variance)

True: More data generally reduces variance because the model becomes less sensitive to the noise in any specific finite training set.

Problem 7 (Nearest Neighbors)

Question: Increasing the value of k used in the k -Nearest Neighbors algorithm will have which of the following impacts on the bias and variance of the model?

- ☐ Bias increases, variance decreases
- ☐ Bias increases, variance remains the same
- ☐ Bias remains the same, variance remains the same
- ☐ Bias decreases, variance increases

Problem 7 (Nearest Neighbors)

Answer: Bias increases, variance decreases

- Increasing k averages over more points, smoothing the decision boundary (lower variance).
- However, it may smooth over local details/patterns, causing the model to underfit (higher bias).

Problem 8 (Cross-Validation)

Question: Suppose you wish to train a ridge regression model using 12-fold cross-validation to choose λ from 10 possible values. How many ridge regression models will you train?

- ☐ 21
- ☐ 241
- ☐ 240
- ☐ 121

Problem 8 (Cross-Validation)

Answer: 121

- For each of the 12 folds, we train 10 models (one for each λ). Total $= 12 \times 10 = 120$.
- After selecting the best λ , we train **one** final model on the entire dataset.
- Total $= 120 + 1 = 121$.

Problem 9 (Naive Bayes)

Question: Which of the following statements are true? (Select all that apply)

- ☐ Naive Bayes is a generative classifier that models the joint distribution $P(X_1, X_2, Y)$.
- ☐ Naive Bayes assumes conditional independence by setting $P(X_1, X_2, Y) = P(X_1)P(X_2)P(Y)$.
- ☐ Laplacian smoothing with parameter $\alpha = 2$ increases the number of parameters that need to be estimated.
- ☐ Classifying a new data point involves finding the class c that maximizes $P(Y = c \mid x_1, x_2)$.

Problem 9 (Naive Bayes)

Answers:

- **True:** Naive Bayes is a generative classifier.
- **False:** The assumption is $P(X_1, X_2|Y) = P(X_1|Y)P(X_2|Y)$, not unconditional independence.
- **False:** Laplacian smoothing changes the count values used in estimation, not the number of parameters.
- **True:** MAP decision rule maximizes the posterior probability.

Problem 10 (Multinomial Logistic Regression)

Question: Which of the following statements are true? (Select all that apply)

- ☐ Multinomial Logistic Regression directly maximizes the fraction of correctly classified data points during training.
- ☐ The cross-entropy loss function is convex; gradient-based optimization finds a global minimum.
- ☐ If $\mathcal{D}$ is linearly separable, Multinomial Logistic Regression will always perfectly classify all training data points.
- ☐ Multinomial Logistic Regression always produces linear decision boundaries.

Problem 10 (Multinomial Logistic Regression)

Answers:

- **False:** It maximizes log-likelihood (a surrogate), not accuracy directly.
- **True:** The objective is convex.
- **False:** Perfect separation implies weights grow to infinity; without regularization or hard constraints, points near the boundary might not be "perfectly" classified in terms of probability (e.g., probability 0.51 vs 0.49).
- **True:** The decision boundaries are defined by $w_j^\top x = w_k^\top x$, which are linear hyperplanes.

Problem 11 (SVM Support Vectors)

Question: Consider a dataset where d_1, d_2, d_3 are class -1 and d_4, d_5, d_6 are class $+1$. The decision boundary is $[2, 0]$ with $b = -2$. Which points are the support vectors?

- ☐ d_3, d_4, d_5
- ☐ d_1, d_2, d_3
- ☐ d_1, d_2, d_4, d_6
- ☐ d_1, d_2, d_6

Problem 11 (SVM Support Vectors)

Answer: d_3, d_4, d_5

- Support vectors are the data points that lie on the margin or within the margin (if soft margin).
- In the provided figure context, these are the points closest to the decision boundary that constrain the margin width.

Problem 12 (SVM Statements)

Question: Which of the following statements about SVMs are true?
(Select all that apply)

- ☐ If training data is linearly separable, hard-margin SVM with linear kernel classifies all training points correctly.
- ☐ If training data is linearly separable, soft-margin SVM with linear kernel classifies all training points correctly.
- ☐ Kernel SVM can learn non-linear decision boundaries.
- ☐ The computational complexity of solving the dual formulation scales sublinearly with the number of features.

Problem 12 (SVM Statements)

Answers:

- **True:** Hard-margin enforces correct classification.
- **False:** Soft-margin allows misclassification to maximize the margin (trade-off).
- **True:** Kernels map to higher dimensions where data might be linearly separable, creating non-linear boundaries in original space.
- **True:** Dual optimization depends on the Gram matrix ($N \times N$), independent of feature dimension d (excluding the kernel computation itself).

Problem 13 (SGD)

Question: Regarding convergence of gradient descent methods:

- ☐ Increasing mini-batch size (fixed LR) yields faster convergence to higher error floor.
- ☐ Increasing learning rate (fixed batch) yields faster convergence to lower error floor.
- ☐ For small enough learning rate η , batch GD exactly converges to solution $\mathbf{w}^*$.
- ☐ The per-iteration computational complexity of SGD and batch GD is the same.

Problem 13 (SGD)

Answer: C

- Batch GD calculates the exact gradient. With a convex objective and appropriate learning rate, it converges to the global optimum.
- SGD oscillates around the optimum due to noise in the gradient estimation.

Problem 14 (Linear Regression GD)

Question: You run GD for linear regression from $\mathbf{w}_0$ to $\mathbf{w}_{100}$. You run it again from $\mathbf{v}_0$ to $\mathbf{v}_{100}$. You find $\mathbf{w}_{100} \neq \mathbf{v}_{100}$. Why?

- ☐ You stopped the algorithm too early (did not converge).
- ☐ The problem has multiple optimal solutions ($X^\top X$ not invertible).
- ☐ You calculated the gradient incorrectly.

Problem 14 (Linear Regression GD)

Answers:

- **Stopped too early:** True. If it hasn't converged, the final point depends on the start point.
- **Multiple optima:** True. If the problem is not strictly convex (feature matrix not full rank), there is a subspace of optimal solutions.

Problem 15 (Poisson MLE)

Question: Given n independent observations $t_1, \dots, t_n$ from a Poisson distribution $P(T = k) = \frac{\lambda^k e^{-\lambda}}{k!}$, what is the MLE for λ ?

- ☐ $\lambda_{MLE} = \frac{\sum t_i}{n}$
- ☐ $\lambda_{MLE} = \frac{n}{\sum t_i}$
- ☐ $\lambda_{MLE} = \sum \frac{1}{t_i}$

Problem 15 (Poisson MLE)

Answer: $\lambda_{MLE} = \frac{\sum_{i=1}^n t_i}{n}$

- Log-likelihood: $\ell(\lambda) = (\sum t_i) \log \lambda - n\lambda - \sum \log(t_i!).$
- Derivative w.r.t λ : $\frac{\sum t_i}{\lambda} - n = 0 \implies \lambda = \frac{\sum t_i}{n}.$

Problem 16 (Logistic Regression)

Context: Dataset: $(x_1, y_1) = (1, 0)$ and $(x_2, y_2) = (0, 1)$. Classifier: $\sigma(w \cdot x)$.

Part A: Suppose we use the standard Log Loss. Show that if we minimize this loss, the optimal value is $w^* = -\infty$.

Problem 16 (Logistic Regression)

Solution Part A: Loss

$\xi(w) = -\log(1 - \sigma(w)) - \log(\sigma(0)) = -\log(\sigma(-w)) - \log(0.5)$. Taking the derivative w.r.t w :

$$\frac{d\xi}{dw} = -\frac{1}{\sigma(-w)}\sigma(-w)(1 - \sigma(-w))(-1) = 1 - \sigma(-w) = \sigma(w)$$

Setting to 0 implies $\sigma(w) \rightarrow 0$, which implies $w \rightarrow -\infty$.

Problem 16 (Logistic Regression)

Part B: Now add L2 regularization: $\frac{\lambda}{2} \|w\|^2$. Let $\lambda = 1$. Is $w^* = -\infty$?

Problem 16 (Logistic Regression)

Solution Part B: No. The derivative becomes:

$$w + \sigma(w) = 0 \implies w = -\sigma(w)$$

Since $\sigma(w) \in (0, 1)$, the solution must be finite (specifically, a negative value where the line $y = w$ intersects $y = -\sigma(w)$).

Problem 16 (Logistic Regression)

Part C & D:

- **C:** If we duplicate the data (4 points total), is $w^* = -\infty$ for the unregularized case?
- **D:** How does increasing λ affect the magnitude of w^* ?

Problem 16 (Logistic Regression)

Solution Part C & D:

- **C:** Yes. The loss is simply doubled. The gradient is just scaled by 2. The condition for stationarity remains $\sigma(w) = 0$, so $w \rightarrow -\infty$.
- **D:** Increasing λ increases the penalty on the norm of w . This pulls the optimal weight closer to 0, decreasing its magnitude.

Problem 17 (Nearest Neighbors)

Context: Test point: $[0.5, 0]^\top$. Training:
 $[1, 0]^\top (l_1)$, $[0, 0.9]^\top (l_2)$, $[-1, 0]^\top (l_3)$, $[0, -1]^\top (l_4)$.

Part A: Using 1-NN (Euclidean), find labels such that test point is “red”.

Problem 17 (Nearest Neighbors)

Solution Part A: Distances to test point $[0.5, 0]$:

- to $[1, 0]$: 0.5 (Closest)
- to others: > 0.5

Since the 1st point is the nearest neighbor, we just need $I_1 = \text{red}$. The others can be anything.

Problem 17 (Nearest Neighbors)

Part B: Distance metric: $d(x, z) = \sqrt{(x_1 - z_1)^2 + \alpha(x_2 - z_2)^2}$. Labels: l_1, l_4 (Red), l_2, l_3 (Blue). Find α such that 1-NN predicts “red”.

Problem 17 (Nearest Neighbors)

Solution Part B: Squared Distances:

- $d_1^2 = 0.5^2 + 0 = 0.25$ (Red)
- $d_2^2 = 0.5^2 + \alpha(0.9^2) = 0.25 + 0.81\alpha$ (Blue)
- $d_3^2 = 1.5^2 + 0 = 2.25$ (Blue)
- $d_4^2 = 0.5^2 + \alpha(1^2) = 0.25 + \alpha$ (Red)

We need $\min(d_1, d_4) < \min(d_2, d_3)$. Since $d_1 = 0.25$, and $d_2 > 0.25$ for any $\alpha > 0$, the closest point is always d_1 (Red). Answer: Any $\alpha > 0$.

Problem 17 (Nearest Neighbors)

Part C: Using 3-NN with same settings. Find α such that prediction is “red” .

Problem 17 (Nearest Neighbors)

Solution Part C: For 3-NN to predict Red (given 2 Red / 2 Blue total points), both Red points (d_1, d_4) must be in the top 3 nearest neighbors. We need d_4 (the farther Red point) to be closer than at least one Blue point (d_2 or d_3).

$$d_4^2 < d_3^2 \implies 0.25 + \alpha < 2.25 \implies \alpha < 2$$

(Note: $d_4^2 < d_2^2$ is impossible for $\alpha > 0$). Answer: $\alpha < 2$.

Problem 18 (SVM Kernel)

Part A: Using dual solution to predict class of $\mathbf{x}$. Solution has s support vectors. How many kernel evaluations?

Problem 18 (SVM Kernel)

Solution Part A: Prediction involves: $f(x) = \sum_{i \in SV} \alpha_i y_i k(x_i, x) + b$. Since $\alpha_i = 0$ for non-support vectors, we only sum over the s support vectors. Answer: s evaluations (to compute the sum) or $2s$ (if including the calculation of b using a support vector).

Problem 18 (SVM Kernel)

Part B: Kernel $k(z_1, z_2) = z_1^\top M z_2$ where $M = \text{diag}(\lambda_1 \dots \lambda_d)$. Find $\phi(z)$.

Problem 18 (SVM Kernel)

Solution Part B: We need $\phi(z_1)^\top \phi(z_2) = z_1^\top M z_2$. Let $\phi(z) = M^{1/2} z$. Since M is diagonal with positive entries, $M^{1/2} = \text{diag}(\sqrt{\lambda_1} \dots \sqrt{\lambda_d})$. Thus: $\phi(\mathbf{x}) = [\sqrt{\lambda_1} x_1, \dots, \sqrt{\lambda_d} x_d]^\top$.

Problem 18 (SVM Kernel)

Part C: Primal SVM variables count given the $\phi(x)$ above.

Problem 18 (SVM Kernel)

Solution Part C: Primal vars: $\mathbf{w}, b, \xi_n$. $\mathbf{w}$ has same dimension as $\phi(x)$, which is d . b is scalar (1). ξ has N entries. Total: $d + 1 + N$.

Problem 18 (SVM Kernel)

Part D: The model misclassified several training points. How to modify SVM to reduce misclassifications?

Problem 18 (SVM Kernel)

Solution Part D: Increase C . A larger C places a higher penalty on the slack variables (ξ), forcing the optimization to prioritize classifying training points correctly over maximizing the margin width.